

Thirty commands, and how to diagnose anything

01

WHERE AM I

<code>pwd</code>	<code>ls -la</code>	here; everything here
<code>which -a python</code>		every match, in PATH order
<code>python -V</code>		version actually running
<code>echo \$PATH</code>	<code> tr ':' '\n'</code>	PATH, one per line
<code>echo \$SHELL</code>		zsh or bash

ENVIRONMENTS

<code>python3.11 -m venv .venv</code>	create
<code>source .venv/bin/activate</code>	enter(mac/Linux/WSL2)
<code>.\.venv\Scripts\Activate.ps1</code>	enter(PowerShell)
<code>deactivate</code>	leave
<code>uv venv --python 3.11</code>	create, fast
<code>uv sync</code>	rebuild from lock; run without
<code>uv run CMD</code>	activating

PACKAGES

<code>python -m pip install X</code>	always <code>-m</code> , never bare <code>pip</code>
<code>python -m pip list</code>	what is installed
<code>python -m pip show -f X</code>	version, location, deps
<code>python -m pip check</code>	broken dependencies
<code>pip index versions X</code>	what versions exist for me
<code>conda env list</code>	conda environments

GIT

<code>git status</code>	run it constantly
<code>git add -p</code> <code>git commit -m</code>	stage by hunk; snapshot
<code>git switch -c feature/x</code>	new branch
<code>git log --oneline --graph -20</code>	recent history, shaped
<code>git restore [--staged] f</code>	discard change; unstage
<code>git reflog</code>	find "lost" work

MACHINE AND GPU

<code>df -h /</code>	<code>du -sh *</code>	free space; what is big
<code>nvidia-smi</code>		driver, VRAM, what is using it
<code>torch.cuda.is_available()</code>		wheel matches driver?
<code>docker system df</code>		where the gigabytes went
<code>tmux new -s job</code>	<code>Ctrl-b d</code>	survive a dropped connection

DIAGNOSE ANYTHING – START AT THE TOP

Impossible / not found / was working yesterday

1. WHICH PYTHON?
`which -a python ; python -V`
`python -c "import sys; print(sys.executable)"`
 ↳ not the one you expected
 → activate, or use the full path
2. WHERE DID IT INSTALL?
`python -m pip show -f PKG`
 ↳ Location outside your env
 → reinstall with `python -m pip`
3. IS THE ENV BROKEN?
`python -m pip check`
 ↳ conflicts
 → delete `.venv`, rebuild from lock
4. IS IT THE NETWORK?
`curl -sSI https://pypi.org/simple/`
 ↳ SSL / proxy / timeout
 → Appendix C, install the root cert
5. IS IT THE GPU STACK?
`nvidia-smi` (driver, on host)
`torch.version.cuda` (None = CPU wheel)
 ↳ mismatch
 → reinstall the `cu###` wheel
6. STILL STUCK?
 Change ONE thing, verify, repeat.
 Then ask, pasting: the command, the full error, `python -V`, which python.

READ THE ERROR, NOT THE PANIC

ModuleNotFoundError	wrong interpreter – step 1
externally-managed-environment	you are in system Python – make a <code>venv</code>
ResolutionImpossible	find the two packages named; relax one
Could not build wheel	no wheel for this Python – change version
CERTIFICATE_VERIFY_FAILED	TLS inspection – Appendix C
Permission denied (publickey)	SSH key not registered
CUDA out of memory	VRAM, not RAM – smaller batch
429 from an API	back off exponentially, add jitter
401 from an API	never retry – check the key

FIVE RULES THAT PREVENT MOST OF IT

1. Change one thing, then verify.
2. One project, one environment, one Python – written down.
3. `python -m pip`, never bare `pip`.
4. If you cannot delete the environment and rebuild it in one command, you do not have one.
5. Secrets live in `.env`, `.env` lives in `.gitignore`, and that line goes in the first commit.